



PROJECT 2

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JUNE 20, 2026
DECODE LABS
Semester (Spring 2026)

Project 2

Phase 1: Univariate Analysis & Distributions

Metric	Count	Mean	Std	Min	25%	Median	75%	Max	Shape Verdict
Quantity	1200	2.95	1.41	1.00	2.00	3.00	4.00	5.00	Skewed Left
UnitPrice	1200	356.41	197.18	11.39	186.06	364.21	521.57	699.93	Skewed Left
ItemsInCart	1200	5.49	2.28	1.00	4.00	5.00	7.00	10.00	Skewed Right
TotalPrice	1200	1053.97	819.86	11.39	410.52	823.62	1578.48	3456.40	Skewed Right

Phase 2: Outlier Diagnostics

- **Quantity:** IQR Method found 0 outliers. Z-Score Method found 0 outliers.
- **UnitPrice:** IQR Method found 0 outliers. Z-Score Method found 0 outliers.
- **ItemsInCart:** IQR Method found 0 outliers. Z-Score Method found 0 outliers.
- **TotalPrice:** IQR Method found 8 potential outliers (Expected bounds: - 1341.41 to 3330.41). Z-Score Method found 0 outliers.

Verdict: The 8 TotalPrice outliers identified by the IQR method represent valid, high-value VIP transactions (Signal, not Noise).

Phase 3: Bivariate Analysis & Mapping Relationships

Pearson Correlation Matrix:

	Quantity	UnitPrice	ItemsInCart	TotalPrice
Quantity	1.00	0.01	0.65	0.62
UnitPrice	0.01	1.00	0.00	0.72
ItemsInCart	0.65	0.00	1.00	0.39

TotalPrice	0.62	0.72	0.39	1.00
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Analysis: UnitPrice has the strongest correlation with TotalPrice ($r = 0.72$).

Phase 4: Executive Summary

1. Problem Statement

Analyze the transactional dataset to understand underlying patterns, distributions, and key drivers of revenue to inform future marketing and sales strategies.

2. Methodology

Executed a forensic Exploratory Data Analysis (EDA) on 1,200 cleaned records. Conducted univariate analysis leveraging the median, deployed the robust IQR method for outlier diagnostics, and mapped relationships using Pearson Correlation Coefficients.

3. Key Findings

Revenue is Right-Skewed: TotalPrice is pulled upward by a minority of high-value purchases (Mean: 1053.97 > Median: 823.61).

Signal, Not Noise: The IQR method unmasked 8 distinct outliers in TotalPrice. These represent valid VIP transactions.

Unit Price > Cart Volume: UnitPrice is the strongest linear predictor of TotalPrice ($r = 0.72$).

4. Recommendations

VIP Investigation: Isolate the 8 outlier transactions to analyze ReferralSource and CouponCode usage.

Strategic Upselling: Shift marketing campaigns toward upselling premium products rather than purely driving higher quantities of cheap items.